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NATIONAL INSTITUTE OF DESIGN

Game Design Document

Line Light

A physical game made using ESP-32 and OpenCV

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Digital Game Design

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Line Light

Overview

Title: Line Light

Engine: OpenCV, Python

Time to design and develop: 6 weeks

Genre: Physical game,

Target Audience: Kids aged from 7-15 years of age.

Team: 2 people

One line description: A movement based multiplayer game where players become lines of light and fight for space by cutting opponents before time runs out.

Description: Light Line transforms body movement into gameplay. Players move physically in space to generate glowing trails on a shared LED grid. By cutting opponents' lines and stealing their length, players compete using movement, timing, and spatial awareness instead of controllers.

Light Line is an interactive real time game that combines camera tracking, computer vision, and LED based visuals to create a shared physical play space. Each player controls a growing line of light through movement, turning the floor into a dynamic battlefield of territory, tension, and reaction. The game focuses on intuitive interaction, readable visuals, and competitive movement driven gameplay.

“What fun can we create with just an LED light?”

That simple question evolved into a massive interactive LED grid where players themselves became lines of light.



Game play visual of three players playing LineLight

Quick Overview

Light Line is an interactive multiplayer installation that transforms physical movement into real time gameplay using light.

Players move inside a camera tracked space, and their movement generates glowing trails on a large LED grid. These trails become the core gameplay element, allowing players to compete, cut opponents, and control space through movement alone.

The project combines computer vision, LED systems, and real time interaction to create a playful physical experience where the floor itself becomes the game.

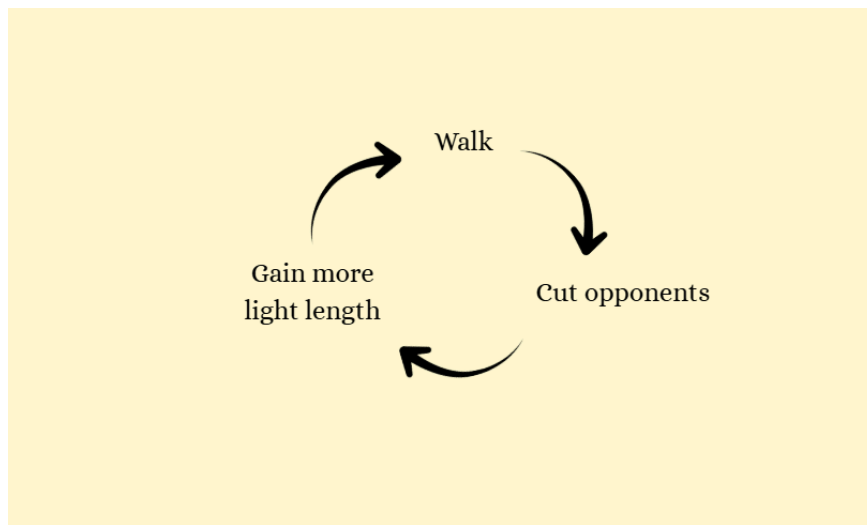
The Core Idea

Imagine becoming a glowing line of light inside a giant living grid.

In Light Line, players physically move through space to create colorful light trails beneath them in real time. These trails are more than visuals. They represent the player's health, territory, and survival inside the arena.

The challenge comes from interaction. Players must dodge, chase, and strategically cut through opponents' trails to steal their length and grow stronger before time runs out. Every movement matters, turning the floor into a fast changing battlefield of light, tension, and movement driven strategy.

Core Loop:



Features:

No Traditional Controllers

Light Line removes traditional controllers entirely. Players interact using their full body movement, making the experience feel immediate, natural, and physically engaging. Instead of pressing buttons, players run, turn, dodge, and chase inside the play space itself.

The Four Es of Interaction

- Embodied

Players embody the line itself. Since movement directly controls the trail, the player and the character begin to feel like one entity rather than separate parts.

- **Embedded**

Holding and moving with the controller creates the feeling of drawing or marking territory on the ground. The interaction feels physical and grounded, almost like walking a glowing pet or carrying a living extension through space.

- **Enactive**

Players do not simply control the game. They enact the role through their body movement. Running, circling, chasing, and escaping become physical performances rather than abstract inputs.

- **Extended**

The controller and glowing trail act as an extension of the player's body. The interaction feels intuitive because the system affords natural movement patterns instead of forcing artificial gestures or commands.

Shared Physical Gameplay Space

Unlike screen based multiplayer games, all players exist together inside the same glowing arena. Every movement leaves behind visible light trails, turning the floor into a constantly evolving battlefield of territory, tension, and interaction.

The game becomes as much about reading other players physically as it is about strategy. Players react not only to the lights on the floor, but also to the movement and behavior of others sharing the space.

Real Time Interactive System

The entire system reacts live through camera tracking, computer vision, and LED feedback. As players move, glowing trails appear instantly beneath them, creating a strong sense of responsiveness and connection between action and feedback.

This makes the installation feel alive rather than pre animated or scripted.

Movement as Gameplay

Movement itself becomes the mechanic. Fast movement can create aggressive attacks, sudden turns can defend territory, and hesitation can make players vulnerable. The gameplay emerges naturally through body motion and spatial awareness rather than complex rules or interfaces.

How It Works



System Pipeline

Camera → Processing → Arduino → LED Grid

Camera Tracking

The experience begins with an overhead camera observing the play space in real time. The camera continuously tracks the position and movement of players as they move across the LED grid. Instead of using traditional controllers, the player's body itself becomes the input system.

The camera acts as the eyes of the project, constantly capturing movement, direction, and spatial interaction between players.

Processing and Game Logic

The camera feed is sent into Processing, where the movement data is interpreted and converted into gameplay behavior.

This stage handles:

- Player position tracking
- Trail generation
- Collision detection
- Line cutting mechanics
- Health and trail length calculations
- Real time game rules and interactions

Rather than directly displaying raw movement, the software transforms player motion into glowing lines, interactive gameplay, and responsive visual feedback.

Arduino Communication

Once the gameplay logic is processed, the information is sent to the Arduino. The Arduino acts as the bridge between software and the physical LED system.

Its role is to:

- Receive updated player and lighting data
- Control LED activation
- Update colors and brightness in real time
- Synchronize visual output across the grid

This allows the physical floor to react instantly to player movement and interaction.

LED Grid Output

The LED grid is the final expressive surface of the experience. Every movement, collision, and interaction is visualized through glowing light trails on the floor.

As players move:

- Trails grow behind them
- Territory becomes visible
- Opponent interactions create cuts and transfers
- The battlefield continuously evolves in real time

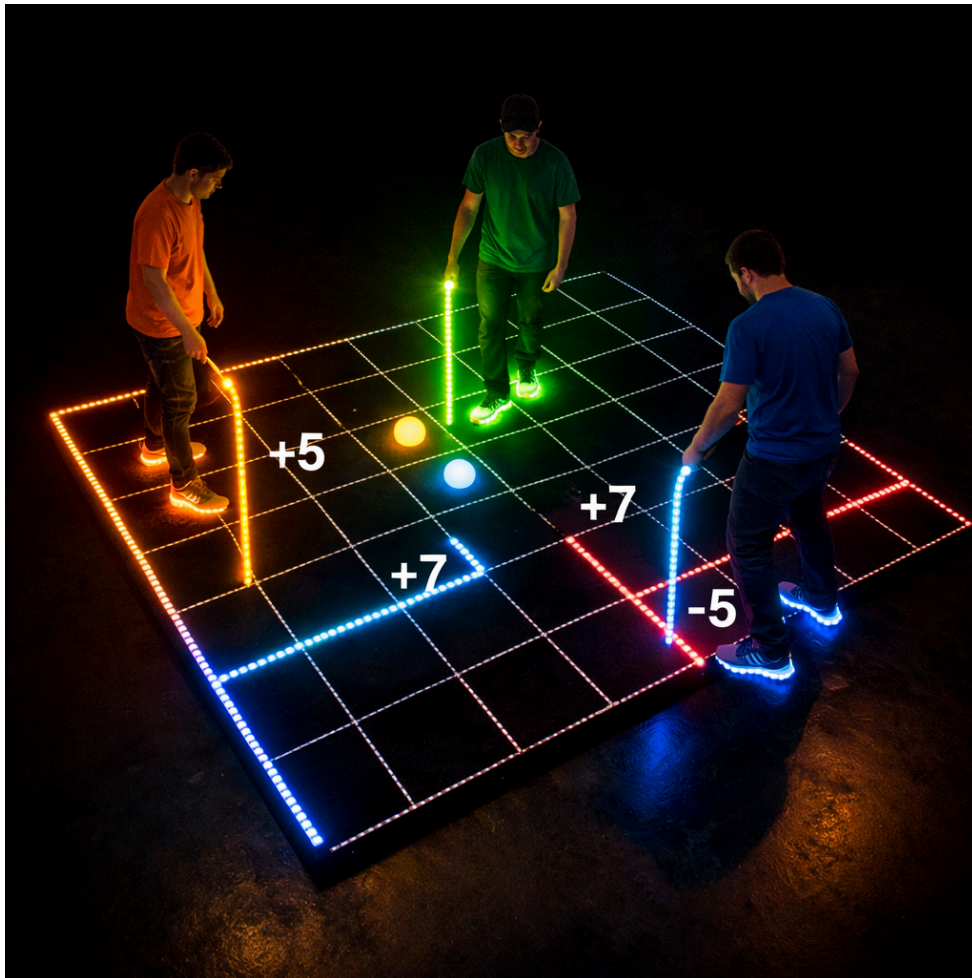
The result is a shared physical game space where movement itself becomes gameplay.

Interaction Loop

The system operates as a continuous feedback loop:

Player Movement → Camera Detection → Game Logic → LED Response → Player Reaction

This loop creates a strong feeling of immediacy and embodiment, making the floor feel alive and responsive to every action.



Initial Representative image

Starting with a simple question

The project began with a very simple question:

“What fun can we create with just a single LED light?”

At first, there was no clear game, no fixed mechanic, and no final vision. We were simply fascinated by the idea of movement interacting with light in real time. The earliest experiments involved turning LEDs on and off based on simple movement detection. Even

seeing a single light react to motion felt exciting because it immediately gave the space a sense of life.

That curiosity slowly expanded into the idea of a grid. Instead of individual LEDs behaving independently, we started imagining a surface that could respond as one large interactive system.

Building the first interactive space

Once the LED grid idea emerged, the focus shifted toward creating a playable physical space. Early versions were extremely rough. LEDs were wired quickly for testing, camera placement was improvised, and movement detection was inconsistent.

At this stage, we were less concerned about polish and more focused on proving the interaction itself. Could movement leave a visible trail? Could multiple people interact on the same surface? Could the floor itself become the game?

The moment trails started appearing behind moving players, the project suddenly became much more than a lighting experiment.

Broken mappings and chaotic movement

One of the most frustrating phases involved LED mapping. The physical wiring order of the LEDs rarely matched the logical layout we imagined in software. A player stepping in one corner could accidentally trigger lights in another section entirely.

At times, the grid felt completely chaotic. Trails appeared mirrored, disconnected, or broken. Small errors in coordinate conversion caused massive visual confusion. We repeatedly tested patterns, walked through the grid tile by tile, and manually corrected mappings until movement finally aligned properly with the physical space.

This phase taught us that interactive systems are deeply dependent on spatial accuracy. Even a small mismatch immediately breaks immersion.

Fighting camera lag and responsiveness

Another major turning point came when we realized how important responsiveness was to the experience. Early camera tracking introduced noticeable delay between movement and light output. The system technically worked, but it did not feel alive.

Fast movement exposed the issue immediately. Players could outrun their own trails, making interactions feel disconnected and unsatisfying.

We spent a significant amount of time experimenting with camera positioning, frame rates, movement smoothing, and processing speed. Slowly, the interaction became tighter and more responsive. Once the lag reduced, the experience transformed completely. Players stopped thinking about the technology and started reacting instinctively to the game.

Learning how brightness affects emotion

Brightness tuning became surprisingly important during development. Initially, the LEDs were pushed to high intensity because brighter visuals seemed more exciting. Instead, the grid quickly became overwhelming and visually exhausting.

The lights felt aggressive rather than playful. Multiple colors bleeding into each other also reduced readability during gameplay.

Through repeated testing, we gradually lowered brightness levels and simplified the visual language. This made the interactions clearer and the space much more comfortable to engage with. We realized that visual restraint often created a stronger experience than maximum intensity.

When mechanics started emerging naturally

Interestingly, the game mechanics were not designed first. They emerged from observing how people interacted with the system.

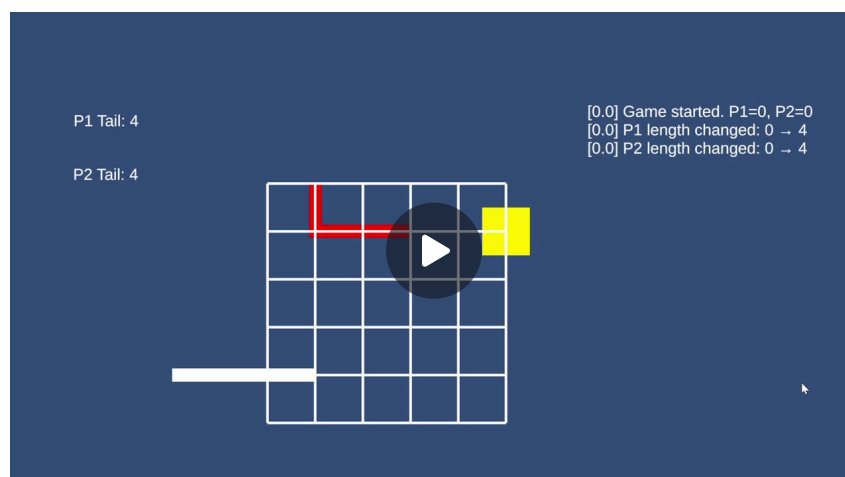
At first, players naturally avoided each other's trails. Then they started intentionally crossing and interfering with them. Soon, people began "attacking" by cutting through paths. Without explicitly designing competition, the grid itself encouraged territorial and strategic behavior.

This became the foundation of the game.

The idea of trails representing health evolved naturally from this interaction. Once cutting mechanics were introduced, the system immediately became more tense and playful. Players became more aware of movement, spacing, and timing.

Over time, additional mechanics such as health transfer, line stealing, timed gameplay, and penalties for crossing one's own trail were added to strengthen this emergent behavior rather than force artificial complexity onto it.

Iteration after iteration



The project evolved almost entirely through iteration. Very few ideas worked perfectly on the first attempt. Most solutions emerged from cycles of testing, breaking, observing, and rebuilding.

Some mechanics looked exciting on paper but became unreadable in practice. Some visual effects distracted from gameplay rather than enhancing it. Certain technical decisions solved one problem while unintentionally creating another.

Instead of treating these moments as failures, we treated them as part of the process. Every broken mapping, delayed response, or chaotic interaction revealed something important about how the system wanted to behave.

From experiment to experience

By the end of development, the project no longer felt like disconnected hardware and software components. It felt like a single responsive system where movement, light, space, and gameplay all supported each other.

What began as a curiosity about LEDs evolved into an embodied multiplayer experience where players physically become glowing lines competing for territory and survival.

The journey was messy, technical, experimental, and constantly changing, but that iterative process ultimately became the strongest part of the project itself.

Challenges and Solutions

Hardware Challenges

Camera Tracking and Detection

One of the earliest challenges was making the camera reliably understand player movement. Small changes in lighting conditions, shadows, and player positioning caused inconsistent tracking. Sometimes movement was detected too late, while at other times the system reacted to noise rather than actual players.

To solve this, the camera placement was repeatedly adjusted to achieve a clearer top down view of the play space. We also refined the detection area and simplified the tracking logic so the system focused only on meaningful movement rather than every visual change inside the frame.

This significantly improved stability and made player movement feel much more reliable.

Frame Rate and Responsiveness

Even when the LEDs updated quickly, the overall experience still felt delayed because the camera frame rate acted as the bottleneck. Fast player movement often outran the visual response, causing trails to appear disconnected from the body.

This became one of the most important discoveries during development. We realized that responsiveness is not only about LED speed but about the entire pipeline working together in sync.

By reducing unnecessary processing, simplifying certain calculations, and balancing camera resolution with frame rate, the interaction became much smoother and more immediate.

LED Brightness and Readability



Initially, the LEDs were operated at very high brightness because we assumed stronger visuals would create a more exciting experience. Instead, the grid became visually exhausting. Colors blended into each other, trails became harder to distinguish, and long play sessions felt uncomfortable.

The solution came through restraint rather than addition. Brightness levels were gradually reduced and colors were simplified to improve readability. Instead of trying to make every LED stand out equally, we focused on making player interactions visually clear.

This made the experience calmer, cleaner, and much easier to understand during gameplay.

Power Distribution and Stability

As the LED count increased, power related issues started appearing across the grid. Certain sections became dimmer than others, especially during moments where multiple colors and trails were active simultaneously.

This revealed the importance of proper power distribution. The issue was solved by improving wiring organization and balancing power delivery across the grid rather than relying on a single source. Once stabilized, the grid behaved much more consistently and visually felt like one unified surface instead of disconnected sections.

Software Challenges

Mapping and Coordinate Translation

One of the most difficult software challenges involved mapping physical player positions to the correct LED locations on the grid. Camera coordinates, game coordinates, and LED indices all existed in different spaces, which caused trails to appear mirrored, shifted, or broken.

Debugging this was especially difficult because the code technically worked while the visuals remained incorrect. We repeatedly tested movement patterns and manually validated grid behavior until the physical and digital spaces aligned correctly.

This iterative refinement became essential for preserving immersion and readability.

Trail Logic and Persistence

Designing the trail system required balancing clarity and gameplay. If trails disappeared too quickly, movement felt meaningless. If they stayed too long, the grid became cluttered and unreadable.

We experimented with fading systems, persistence timing, and dynamic trail behavior before deciding to prioritize stronger persistence and visible territory. This made movement feel consequential and reinforced the idea that players were actively shaping the space around them.

Performance Optimization

As more mechanics were introduced, performance became a growing concern. Updating large numbers of LEDs while simultaneously processing camera input and interaction logic created noticeable slowdowns.

The initial approach updated too many values every frame, which caused stuttering during gameplay. To solve this, we optimized the update flow by reducing unnecessary calculations

and limiting updates only to regions actively changing.

This significantly improved responsiveness without reducing visual quality.

Interaction and Collision Systems

The interaction system evolved naturally but became increasingly complex over time. Cutting mechanics, line ownership, health transfer, and penalties for crossing one's own trail all introduced edge cases that often behaved unpredictably.

At certain points, interactions became chaotic because multiple collisions could happen simultaneously across the grid. Solving this required restructuring the interaction logic so collisions were handled consistently and clearly.

Over time, the system shifted from reactive behavior to rule based interaction, making gameplay feel fair, readable, and intentional instead of random.

Iteration as the Solution

Most challenges in this project were not solved instantly. Nearly every system required multiple rounds of testing, failure, and refinement before reaching a stable state.

Rather than treating iteration as a setback, it became the core development method of the project. Each problem revealed new insights about how hardware, software, and gameplay needed to support each other as one connected experience.

8. Game Design Evolution

Iteration 1

Movement becomes Light

The earliest version of Light Line had no competition, no scoring system, and no rules. Players simply moved through the space while glowing trails appeared beneath them in real time.

What surprised us was how naturally people began interacting with the grid. Some players tried avoiding existing trails while others intentionally crossed through them. Even without mechanics, the system already created a sense of movement, territory, and ownership.

This was the moment we realized the interaction itself was already fun.

The grid no longer felt like a lighting setup. It started feeling like a playable space.

Iteration 2

The Birth of Conflict

The second major iteration introduced the cutting mechanic.

Players could now cross through opponents' trails and remove parts of their line. This instantly transformed the experience from passive interaction into competition. Movement suddenly became strategic.

Fast movement created aggressive attacks. Sharp turns became defensive maneuvers. Players started predicting each other's movement and intentionally trapping opponents.

The floor became less of a display and more of a battlefield.

However, this version quickly became chaotic. Since cutting only punished opponents without rewarding the attacker, gameplay often turned into pure destruction. Trails disappeared too quickly, and players lacked motivation to survive intelligently.

This forced us to rethink balance and consequence.

Iteration 3

Health Transfer and Survival

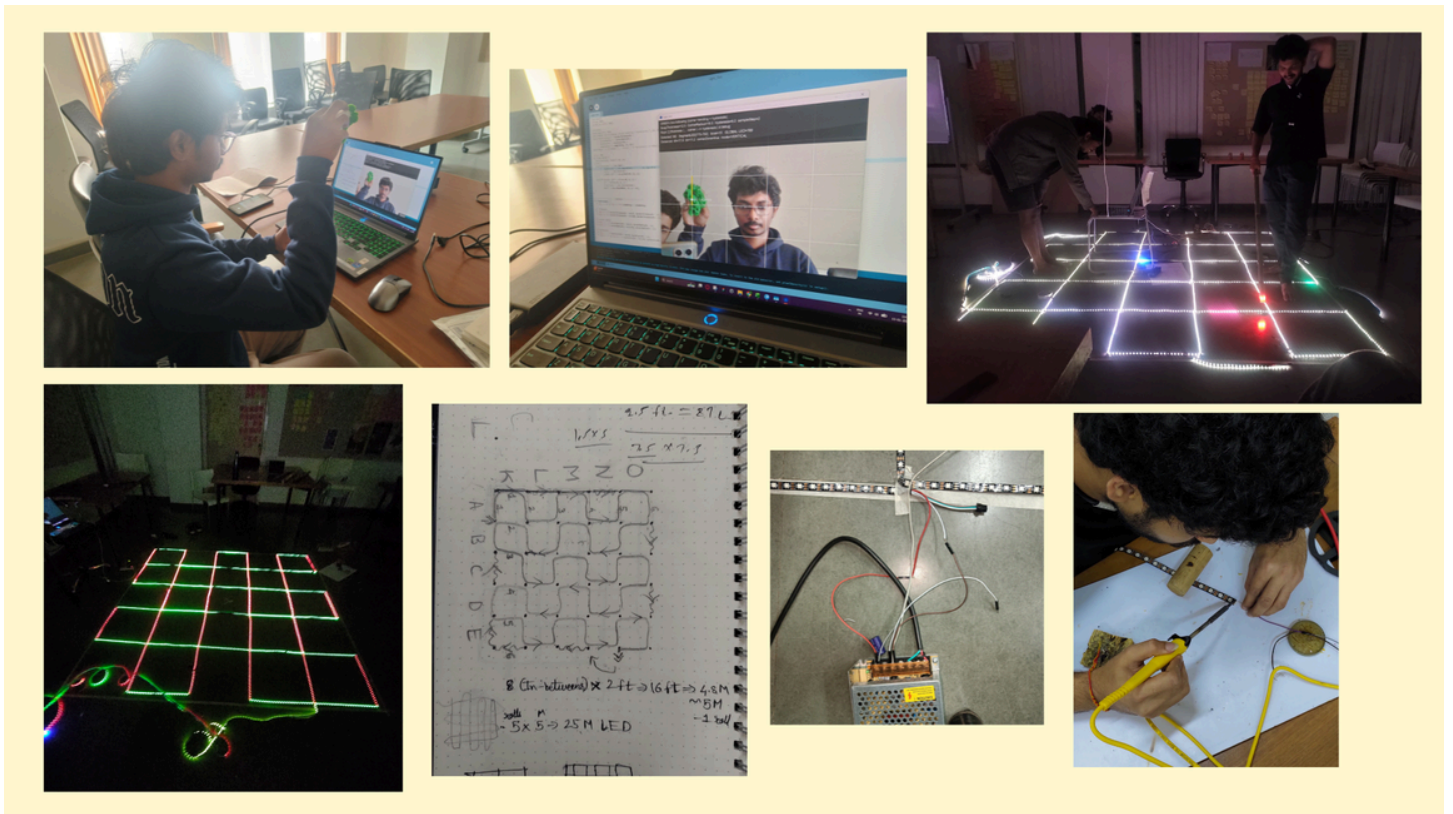
The final major iteration refined the cutting system by introducing health transfer.

Now, when a player cut another player's line, the removed section transferred to the attacker instead of simply disappearing. This made every interaction meaningful and introduced a clear risk versus reward dynamic.

At the same time, timed gameplay was added. Players now had a limited duration to survive, expand, and dominate the arena before the match ended.

The game evolved from playful movement into a tense physical competition.

Players became more aware of spacing, timing, and territorial control. Every movement mattered because every mistake had visible consequences on the grid.



9. Final Experience

What the Player Feels

Light Line is designed to make players feel physically present inside the game rather than controlling it from outside.

The moment players step onto the grid, their movement leaves behind glowing trails of light. Every step creates visible consequences. Every turn reshapes the arena.

The floor remembers movement.

As the game progresses, the space slowly fills with tension, territory, and interaction. Trails become visual memories of player behavior. Some areas become dangerous, others become safe zones, and the grid itself starts telling the story of the match.

Movement Driven Gameplay

There are no buttons or joysticks. The body itself becomes the controller.

Players attack by running. Defend by changing direction. Escape by reading space. The gameplay emerges entirely through natural movement and physical awareness.

Because of this, interactions feel instinctive rather than mechanical.

Players are not controlling characters. They become the moving entities themselves.

Tension and Territory

The game constantly creates pressure without relying on complex systems.

Standing still feels risky. Moving too aggressively can expose a player's trail. Every interaction changes the state of the arena for everyone sharing the space.

The glowing lines begin feeling less like visuals and more like extensions of the players themselves.

This creates a strong emotional connection between movement, survival, and territory control.

A Physical Multiplayer Experience

Unlike traditional screen based multiplayer games, Light Line exists in shared physical space. Players react not only to the LED grid but also to each other's real movement and body language.

This creates moments of:

- Chasing
- Bluffing
- Cornering
- Escaping
- Territory claiming
- Sudden tension

The experience feels closer to a physical sport made out of light.

10. Reflection and Takeaways

Technical Learnings

This project taught us how deeply connected hardware, software, and interaction design are within real time systems.

Small hardware changes completely affected gameplay feel. Camera responsiveness influenced player movement. Brightness affected readability. LED update speed affected tension and immersion.

We learned how to build a complete interactive pipeline connecting:

Camera Tracking → Processing → Arduino → LED Grid

Every stage affected the final player experience.

Understanding System Design

One of the biggest takeaways was understanding that interactive systems are not built perfectly from the start. They evolve through iteration.

Many mechanics that seemed exciting initially became confusing during testing. Some visual effects looked impressive technically but reduced gameplay clarity. Simpler solutions often created stronger player experiences.

The project taught us to design through observation rather than assumption.